

Wireshark Traffic Analysis

Cybersecurity Portfolio Project

1. Project Overview

This project focused on capturing and analyzing network traffic using Wireshark. The lab demonstrated how network packets can be monitored to observe FTP authentication traffic, user login activity, and communication between systems on a local network.

2. Objective

The goal of this project was to analyze network traffic generated during an FTP login session and identify communication details such as protocols, source and destination IP addresses, and authentication activity.

3. Tools Used

- Kali Linux
- Wireshark
- FileZilla Server
- FTP Protocol
- VirtualBox

4. Skills Demonstrated

- Packet Analysis
- Network Traffic Monitoring
- Protocol Analysis
- FTP Traffic Inspection
- Security Monitoring
- Wireshark Usage
- Network Troubleshooting

5. Lab Environment

The lab environment consisted of:

- A Kali Linux virtual machine
- A Windows 10 virtual machine running FileZilla Server
- A local virtual network configured inside VirtualBox

Wireshark was used to capture and analyze network traffic generated during FTP communication between the systems.

6. Steps Performed

1. Configured FileZilla Server on the Windows virtual machine.
2. Created FTP user accounts and authentication settings.
3. Connected to the FTP server from Kali Linux using the FTP client.
4. Logged into the FTP server using test credentials.
5. Captured network traffic using Wireshark.
6. Filtered and analyzed FTP packets.
7. Observed authentication and login traffic between systems.

7. Screenshots

Figure 1 — Configuring FileZilla Server Listener Settings

- Configured FTP server listener settings.
- Verified server communication ports.

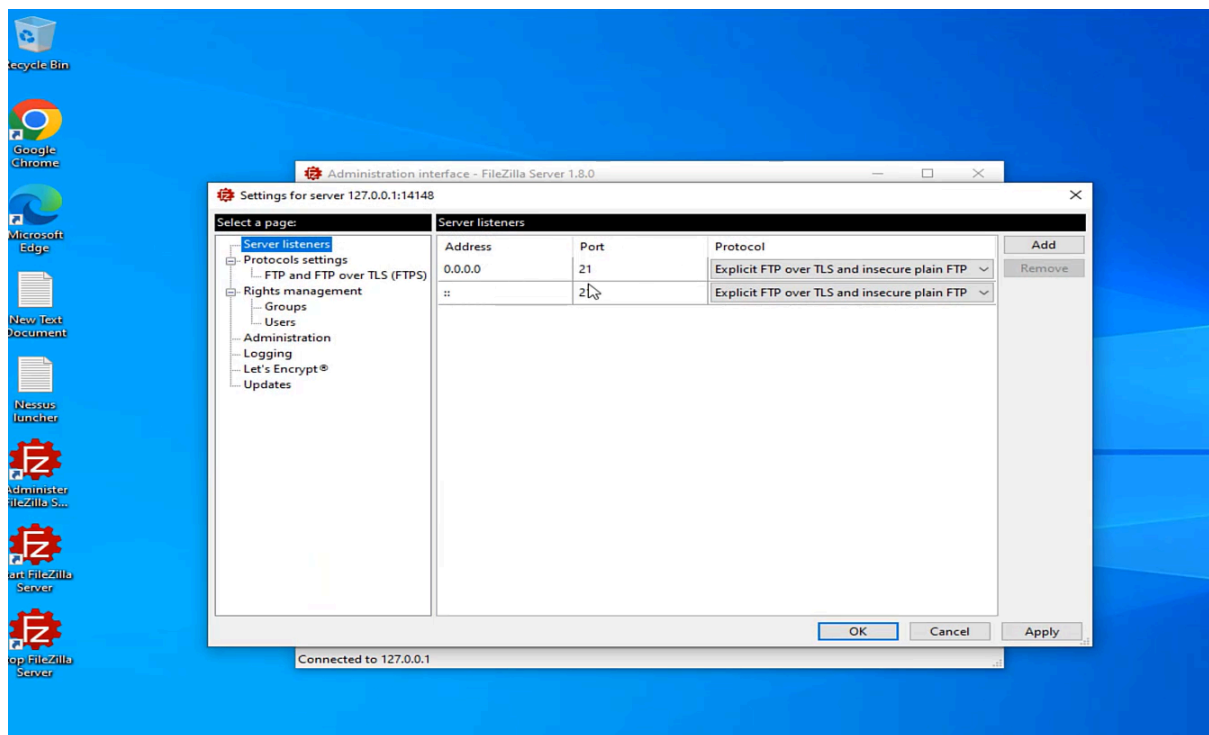


Figure 2 — Managing FTP User Accounts

- Created and managed FTP user accounts.
- Configured authentication settings.

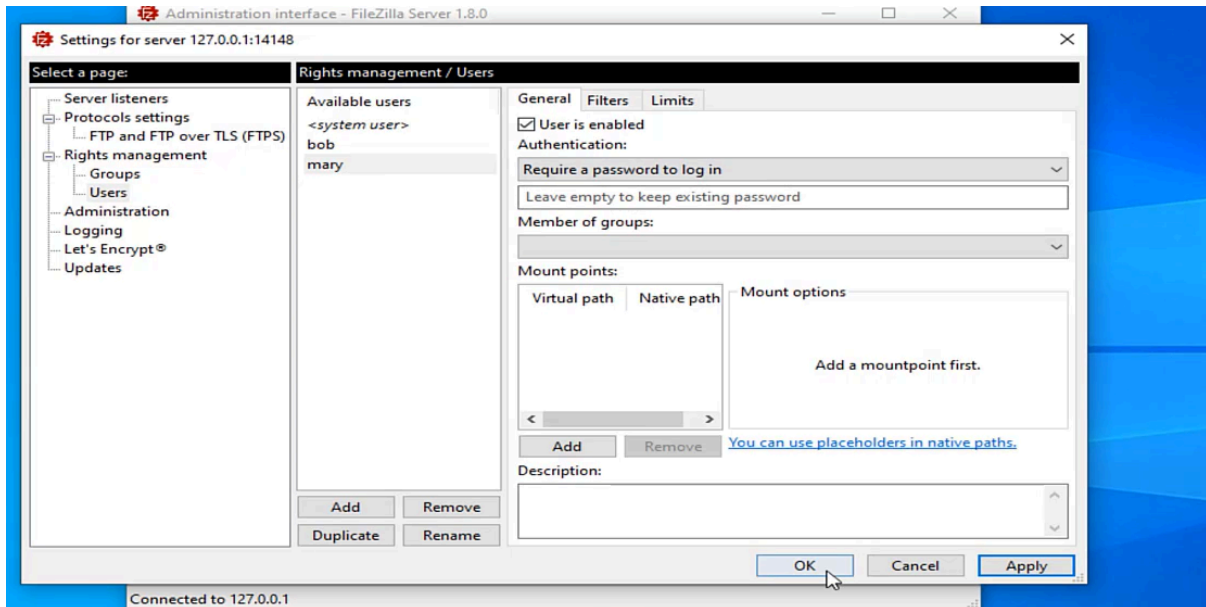


Figure 3 — FTP Login Session from Kali Linux

- Connected to the FTP server using Kali Linux.
- Successfully authenticated to the FileZilla server.

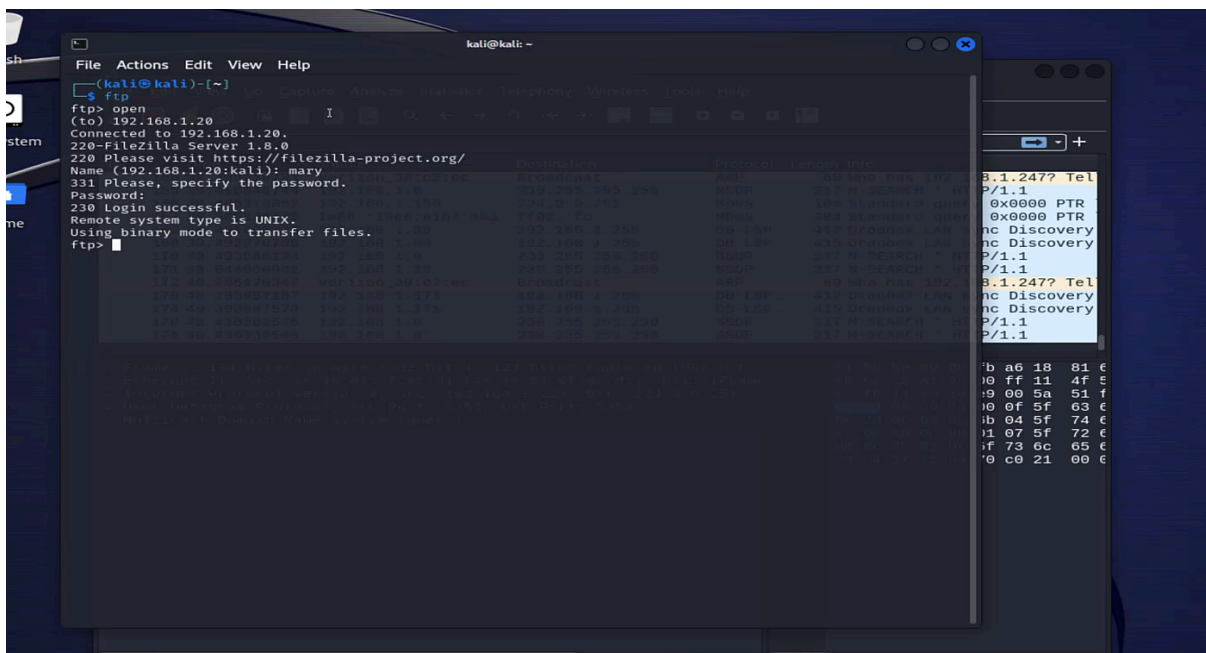
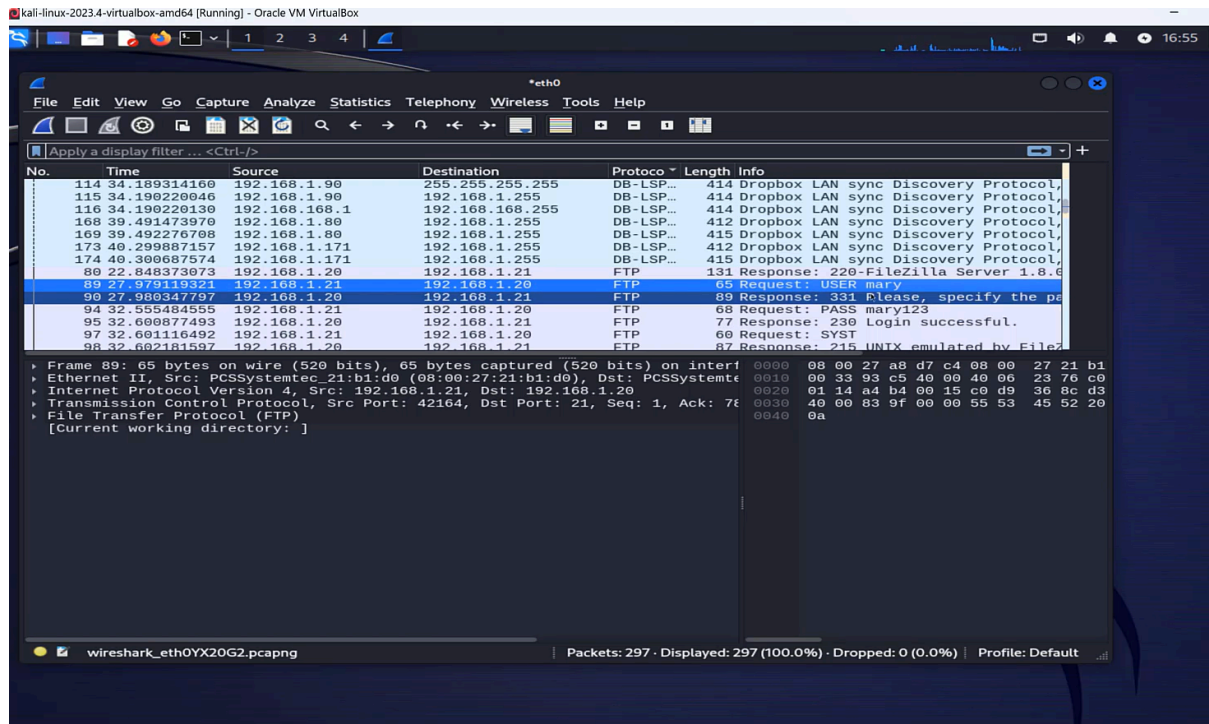


Figure 4 — Wireshark FTP Packet Analysis

- Captured FTP authentication traffic.
- Observed FTP protocol communication.
- Analyzed packet details including:
 - source IP
 - destination IP
 - FTP login requests
 - server responses



8. Security Benefits

Network traffic analysis helps security teams:

- Detect suspicious activity
- Monitor authentication attempts
- Investigate incidents
- Identify insecure protocols
- Troubleshoot network communication

This project also demonstrated how unencrypted FTP traffic can expose authentication activity on a network.

9. What I Learned

I learned how Wireshark captures live network traffic and how protocol analysis can reveal communication details between systems.

I also gained experience analyzing FTP authentication traffic and identifying packet-level network activity.

10. Conclusion

This project demonstrated practical cybersecurity skills related to network monitoring, packet analysis, and traffic inspection using Wireshark in a virtual lab environment.